

AIYSHAH MEERAB

Computer Science Student

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ABOUT ME

Detail-oriented Computer Science student passionate about coding and innovation. Looking to leverage academic knowledge and practical skills through a challenging internship opportunity.

EDUCATION

FAST-NU Lahore

2024 – 2028

Bachelor of Science in Computer Science — GPA: 3.24 (Fourth Semester)

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, Assembly (16-bit)

Core Strengths: Communication, Team Work, Attention to Detail

PROJECTS

Deaf & Blind Assistant | Python, OpenCV, YOLO, TensorFlow

- Trained and fine-tuned object-detection and hand-gesture recognition models, achieving 99% recognition accuracy by tracking performance via confusion matrices and accuracy/loss line graphs to diagnose and correct overfitting.
- Applied transfer learning on a YOLO backbone with data augmentation and IoU-based non-max suppression to improve bounding-box precision across varied lighting conditions.
- Built a real-time computer-vision pipeline for object and hand-gesture detection to assist visually and hearing impaired users, developed collaboratively under supervisor Mr. Usman Anwer.

Unifi / NU Connect | React Native, Expo, FastAPI, SQLAlchemy

- Designed the app's UI/UX in Figma, translating wireframes into a polished, user-friendly interface.
- Implemented token-based authentication and role-based admin controls for content moderation and post editing.
- Architected a normalized relational schema via SQLAlchemy ORM with RESTful FastAPI endpoints for scalable CRUD operations.
- Built real-time direct messaging and a followers/following system alongside a live feed with posts, likes, and comments.

Snake Game | C++

- Designed 3 distinct gameplay modes, including a customizable obstacle course mode allowing players to place and rearrange obstacles to reshape difficulty and navigation on the fly.
- Applied object-oriented design principles and a fixed-timestep game loop to manage entity state and rendering cycles.
- Integrated background music and sound effects alongside collision detection and live score tracking.

Pacman | C++

- Programmed distinct tracking techniques per ghost to trail Pacman, varying difficulty across characters.
- Utilized finite-state machines and grid-based pathfinding logic to govern ghost behavior transitions in real time.
- Implemented a power-pellet mechanic triggering a temporary vulnerable/invisible mode for ghosts.

Balloon Pop | x86 Assembly (16-bit)

- Hand-rendered every graphic element pixel-by-pixel via direct video memory manipulation, with a restart function to reset game state.
- Managed low-level memory addressing and interrupt handling to optimize rendering cycles within limited system resources.
- Debugged the 16-bit assembly game loop at the instruction level to synchronize input, rendering, and timing for smooth, flicker-free gameplay.

CERTIFICATIONS

- Deloitte (Forage)** — AI, Technology & Cybersecurity Job Simulations | 2026
- DevTown** | 2026 — AI Chatbot
- Google Developer Groups - VIT-AP × Microsoft Student Chapter (MSIT)** — Operating Systems Bootcamp, Project Completion | Feb 2026
- HP** | 2025 — Introduction to Cybersecurity Awareness; AI for Beginners
- University of Helsinki** | 2023 — Elements of AI
- Open Weaver** | 2023 — Basics of Python Programming
- WHO** | 2023 — Ethics & Governance of AI for Health
- LUMS** | 2022 — Winter School

EXPERIENCE

SOFTEC — FAST-NU | Team Lead

2025

SOFTEC — FAST-NU | Media Team & Query Vista Officer

2026